Task 7

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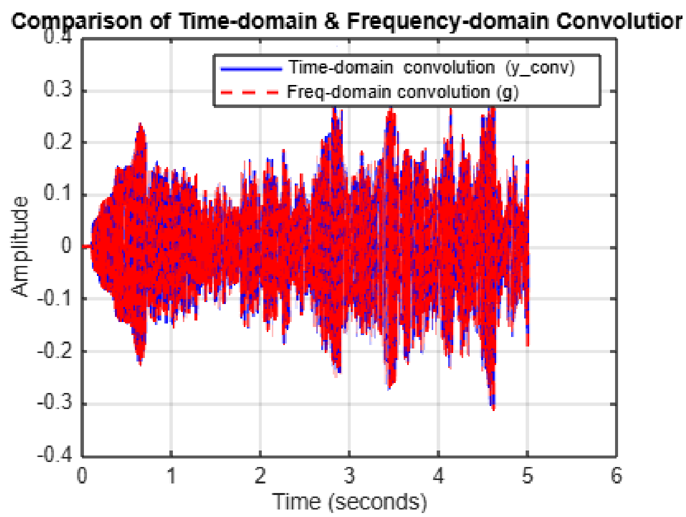
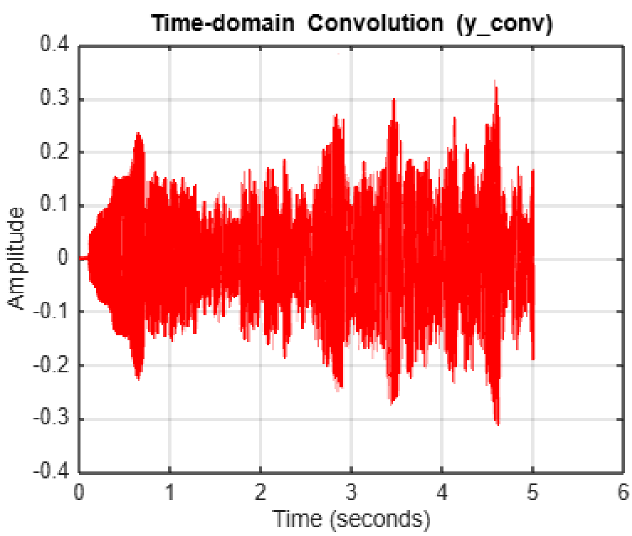
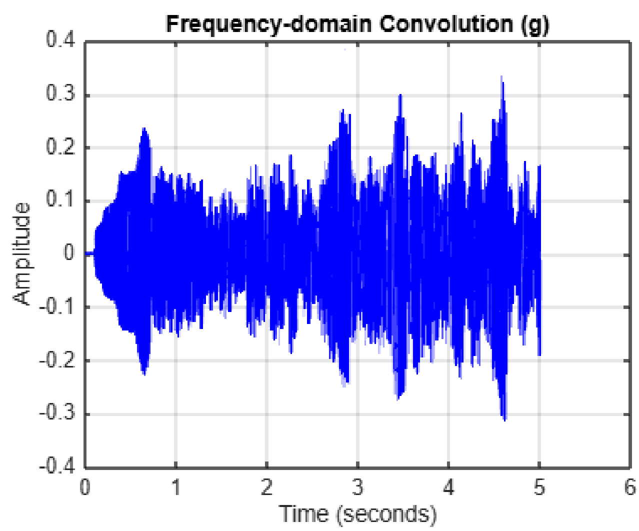
C.

Are they identical (within a small numerical error)?

This is the plot output:







Yes, they are mathematically equivalent and should produce the same result within a very small numerical error. The time-domain convolution directly calculates the convolution sum, while the frequency-domain convolution uses the FFT and IFFT based on the Convolution Theorem.

Small differences between the two results arise due to numerical precision limits in floating-point calculations, especially from the FFT and IFFT steps. Additionally, the frequency-domain result may have tiny imaginary components caused by numerical inaccuracies, which can be ignored by taking the real part.

Explain why this theorem is so important for applying filters efficiently to long signals.

The Convolution Theorem tells us that convolution in the time domain is equivalent to multiplication in the frequency domain. This is significant because:

1. Computational Efficiency:

   Convolution of a long signal with a filter in the time domain directly comprises a number of operations equal to the product of their lengths — which may be enormously high and slow.

2. Faster Processing via FFT:

   By transforming both the signal and the filter to the frequency domain through the Fast Fourier Transform (FFT), convolution is simply a point-wise multiplication of their spectra. FFT and inverse FFT operations take $O(N \log N)$ time, which is significantly less than the $O(N^2)$ operations of direct convolution for large N.

3. Practicality in Actual Applications for Extremely Long Signals

   For real-time streaming data or long signals, application of the frequency domain technique makes filtering computationally viable and efficient, and it enables programs such as audio processing, image filtration, and communication systems to run efficiently and in real-time.

In summary, the Convolution Theorem allows us to replace time-domain convolution with extremely fast frequency-domain multiplication, which makes filtering of long signals efficient and viable.